

Assignment - 1

(1) (a) line segment, $A(2, 5)$
 $B(-2, -5)$

$$dx = -2 - 2 = -4$$

$$dy = -5 - 5 = -10$$

$$|dy| > |dx| \quad ; \quad \text{zone} = 5$$

converting ^{the} end points to zone 0;

$$A(2, 5) \rightarrow A'(-5, -2)$$

$$B(-2, -5) \rightarrow B'(5, 2)$$

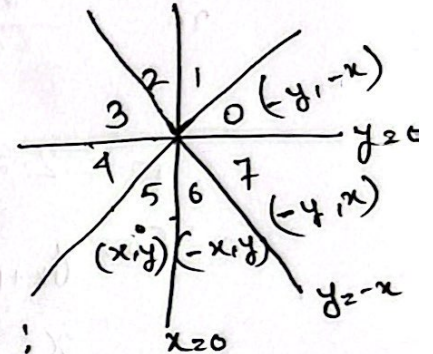
$$dx' = 5 + 5 = 10$$

$$dy' = 2 + 2 = 4$$

$$D = 2 \times 4 - 10 = -2$$

$$\Delta NE = 2 \times 4 - 2 \times 10 = -12$$

$$\Delta E = 2 \times 4 = 8$$



x'	y'	D	NE E	$d(\text{updating})$	x	y	pixel
-5	-2	-2	E	6	2	5	(2, 5)
-4	-2	6	NE	-6	2	4	(2, 4)
-3	-1	-6	E	2	1	3	(1, 3)
-2	-1	2	NE	-10	1	2	(1, 2)
-1	0	-10	E	-2	0	1	(0, 1)
0	0	-2	E	6	0	0	(0, 0)
1	0	6	NE	-6	0	-1	(0, -1)
2	1	-6	E	2	-1	-2	(-1, -2)
3	1	2	NE	-10	-1	-3	(-1, -3)
4	2	-10	E	-2	-2	-4	(-2, -4)
5	2	-2	E	6	-2	-5	(-2, -5)

(b) Here,

$$A(-2, -5)$$

$$B(2, 5)$$

$$m = \frac{5+5}{2+2} = \frac{10}{4} = 2.5$$

$m = 2.5$, it doesn't satisfy $-1 < m < 1$ condition.

$$\text{So, } y_{k+1} = y_k + 1$$

$$x_{k+1} = x_k + \frac{1}{m}$$

$y(+1)$	$x(+1/m)$	$x(\text{roundoff})$	pixel
-5	-2		
-4	-1.6	-2	$(-2, -4)$
-3	-1.2	-1	$(-1, -3)$
-2	-0.8	-1	$(-1, -2)$
-1	-0.4	0	$(0, -1)$
0	0	0	$(0, 0)$
1	0.4	0	$(0, 1)$
2	0.8	1	$(1, 2)$
3	1.2	1	$(1, 3)$
4	1.6	2	$(2, 4)$
5	2	2	$(2, 5)$

(2) (a) resolution = 2560×1440
frame rate = 75 fps

Total number of pixels in a single frame =
$$\frac{2560 \times 1440}{1} = 3686400 \text{ pixels}$$

(b) Time to generate 1 frame = $\frac{1}{75} = 0.0133 \text{ s}$
 $= 13.33 \text{ ms}$

(c) GPU can process 75000 pixels per ms.

from (a), total number of pixels = 3686400 pixels
in a single frame

The time to render one entire frame:

$$\frac{3686400}{75000} \text{ ms}$$
$$= 49.152 \text{ ms}$$

From (b), we see that to maintain 75 fps,
^{should} GPU ~~take~~ 13.33 ms to generate one frame.

However, from (c), we found that GPU takes
49.152 ms to generate one frame. So, it
cannot render one entire frame within the
required time to maintain 75 fps.